OLA Ride Booking Analysis Report

Executive Summary

This project focuses on analyzing OLA ride booking data using Google Sheets, SQL, and Power BI to uncover valuable business insights related to ride demand, booking performance, customer behavior, cancellations, revenue generation, and service quality. The dataset was initially cleaned and prepared using Google Sheets, analyzed using SQL, and visualized through interactive Power BI dashboards. The dashboard provides a comprehensive view of operational performance and helps identify areas for business improvement.

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1. Project Objective

The primary objective of this project is to analyze ride booking data and generate actionable insights that can help improve customer satisfaction, operational efficiency, and overall business performance.

Key objectives include:

- Monitoring ride booking trends
- Evaluating booking success and cancellation rates
- Analyzing revenue generation
- Understanding customer and driver behavior
- Identifying major cancellation reasons
- Assessing service quality through ratings

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2. Dataset Overview

The dataset contains approximately **103,024 ride bookings** for July 2024.

Key Attributes

- Booking ID
- Date & Time
- Customer ID
- Vehicle Type
- Booking Status
- Pickup & Drop Location
- Booking Value
- Ride Distance
- Payment Method
- Driver Rating
- Customer Rating
- Cancellation Reasons

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3. Tools & Technologies Used

Google Sheets

Used for:

- Data cleaning and preprocessing
- Handling missing and inconsistent values
- Initial data exploration
- Preparing the dataset for SQL analysis and Power BI visualization

SQL

Used for:

- Data extraction
- Data analysis
- Aggregations and filtering
- Creating analytical views

Power BI

Used for:

- Data visualization
- Dashboard development
- KPI tracking
- Interactive business reporting

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4. Key Performance Indicators (KPIs)

KPI	Value
Total Bookings	103,024
Total Booking Value	₹35 Million+
Successful Bookings	63,967
Success Rate	62.09%
KPI	Value
Cancelled Bookings	28,933
Cancellation Rate	28.08%

5. SQL Analysis Performed

The following business questions were answered using SQL:

1. Retrieve All Successful Bookings

2. Calculate Average Ride Distance by Vehicle Type

Vehicle_Type	Avg_Distance
Prime Sedan	15.7649
Bike	15.5331
Prime SUV	15.2745
eBike	15.5806
Mini	15.5101
Prime Plus	15.4475
Auto	6.2381

3. Count Customer-Cancelled Rides

count(*)
10499

4. Identify Top 5 Customers Based on Ride Frequency

Customer_ID	Total_Rides
CID954071	5
CID539191	4
CID189965	4
CID268274	4
CID952434	4

5. Analyze Driver Cancellations Due to Personal and Vehicle-Related Issues

count(*)
6542

6. Find Maximum and Minimum Driver Ratings for Prime Sedan Rides

Max_Rating	Min_rating
5	3

7. Calculate Average Customer Ratings by Vehicle Type

Vehicle_Type	Average_Ratings
Prime Sedan	4
Bike	3.99
Prime SUV	4
eBike	3.99
Mini	4
Prime Plus	4.01
Auto	4

8. Compute Total Revenue from Successful Rides

total_successfull_value
35080467

6. Dashboard Analysis & Insights

1. Ride Volume Over Time

The ride volume remained relatively stable throughout the month, with slight increases on certain dates indicating periods of higher demand.

Insight:

Demand forecasting can help allocate drivers more efficiently during peak periods.

6.2 Booking Status Analysis

Booking outcomes were categorized into:

- Successful Rides
- Cancelled by Customer

- Cancelled by Driver
- Driver Not Found **Insight:**

Although the success rate exceeded 62%, nearly one-third of bookings were unsuccessful, indicating opportunities for operational improvement.

6.3 Revenue Analysis

The platform generated over **₹35 million** in booking value during the analysis period.

Insight:

Revenue performance demonstrates strong platform usage and customer demand.

6.4 Payment Method Analysis

Revenue was distributed across:

- Cash
- UPI
- Credit Card
- Debit Card

Insight:

Digital payment methods, particularly UPI, contribute significantly to transactions, highlighting customer preference for cashless payments.

6.5 Top Customers Analysis

The dashboard identified the top five customers based on total booking value.

Insight:

These high-value customers can be targeted with loyalty programs, rewards, and personalized offers to improve retention.

6.6 Customer Cancellation Analysis

Major customer cancellation reasons include:

- Driver not moving toward pickup location
- Driver asked to cancel
- Change of plans

- AC not working
- Wrong address **Insight:**

Driver-related issues account for a substantial portion of customer cancellations, affecting customer experience.

6.7 Driver Cancellation Analysis

Top driver cancellation reasons include:

- Personal and vehicle-related issues
- Customer-related issues
- Customer illness concerns
- Excess passengers **Insight:**

Vehicle maintenance and driver support initiatives could significantly reduce driver cancellations.

6.8 Ratings Analysis

Average Driver Rating

~4.0/5

Average Customer Rating

~4.0/5

Insight:

Both ratings indicate a generally positive service experience and consistent service quality.

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7. Business Recommendations

Reduce Cancellation Rate

- Improve driver allocation efficiency.
- Monitor driver movement toward pickup locations.
- Introduce penalties for unnecessary cancellations.
- Enhance real-time communication between drivers and customers.

Improve Customer Satisfaction

- Conduct regular vehicle quality inspections.
- Ensure proper air conditioning and vehicle maintenance.
- Improve pickup location accuracy.

Increase Revenue

- Launch targeted offers for frequent riders.
- Promote digital payment incentives.
- Introduce loyalty and subscription programs.

Improve Driver Performance

- Conduct driver training programs.
- Offer performance-based incentives.
- Provide better support for vehicle-related issues.

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8. Conclusion

The OLA Ride Booking Analysis project successfully leveraged Google Sheets, SQL, and Power BI to analyze booking performance, customer behavior, cancellations, revenue, and service quality.

The analysis revealed that while the platform maintains a healthy booking success rate of over 62%, reducing cancellations presents a significant opportunity to improve operational efficiency and customer satisfaction. The dashboard provides actionable insights that can support data-driven decision-making and business growth.

Overall, this project demonstrates practical skills in data cleaning, data preprocessing, SQL querying, data analysis, KPI development, dashboard design, and business intelligence reporting.